



Creating a Pipeline using Airflow UI & Docker

Apache Airflow



Apache Airflow is an open-source platform that allows you to programmatically create, schedule, and monitor workflows.

- “Programmatically” means you define workflows using code (usually Python) rather than manually clicking in a UI.
 - Workflow refers to a series of tasks that need to run in a specific order (like ETL jobs, data processing, or ML pipelines).
 - Airflow provides tools for dependency management, scheduling, retries, logging, and monitoring of these workflows.
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Key Components:

- DAG (Directed Acyclic Graph): Defines workflow and task order
- Task: Single unit of work (Python function, Bash script, SQL query)
- Scheduler: Triggers tasks as per schedule
- Executor: Runs tasks (Local, Celery, Kubernetes)
- Web UI: Monitor pipelines, view logs, manage DAGs

Docker

Docker is an open-source platform that packages applications and their dependencies into lightweight containers.

These containers run consistently across any environment, making deployment and scaling easy.

Docker Concept

- Container – A lightweight, isolated unit that runs an application.
- Image – A read-only template used to create containers.
- Dockerfile – Script to build a Docker image.
- Docker Compose – Tool to define and run multi-container applications.
- Volume – Persistent storage for data inside containers.
- Registry – Repository to store and share images (e.g., Docker Hub).

Airflow ETL Workflow

1. Extract (S3)

- Task: Download data from Amazon S3 bucket
- Operator: S3Hook or PythonOperator

2. Transform

- Task: Clean, format, or aggregate data
- Operator: PythonOperator

3. Load (PostgreSQL)

- Task: Insert transformed data into PostgreSQL database
- Operator: PostgresOperator or PythonOperator with psycopg2